

Exponentials and logarithms - Set 2 - Solutions

Specification MM5.1-MM5.3.

Answer key

Question	Answer	Topic
1	E	Exponentials and logarithms
2	B	Exponentials and logarithms
3	A	Exponentials and logarithms
4	B	Exponentials and logarithms
5	A	Exponentials and logarithms
6	E	Exponentials and logarithms
7	D	Exponentials and logarithms
8	E	Exponentials and logarithms
9	B	Exponentials and logarithms

Worked solutions

Question 1 LOG-16 · Exponentials and logarithms · spec MM5.1

For which of the following values of x is $3^x < 2^{x+1}$?

I $x = 0$

II $x = 1$

III $x = 2$

Answer: E

$x = 0$: $1 < 2$ true. $x = 1$: $3 < 4$ true. $x = 2$: $9 < 8$ false. So I and II.

Question 2 LOG-18 · Exponentials and logarithms · spec MM5.2

If $\log_a b = 2$ and $\log_b c = 3$, find $\log_a c$.

Answer: B

$b = a^2$ and $c = b^3 = a^6$, so $\log_a c = 6$.

Question 3 LOG-06 · Exponentials and logarithms · spec MM5.1

The graph of $y = 2^x$ is translated 3 units in the positive x -direction. The resulting graph has equation

Answer: A

$y = 2^{x-3} = 2^{-3} \cdot 2^x = \frac{1}{8} \cdot 2^x$. (A translation of an exponential is also a stretch.)

Question 4 LOG-08 · Exponentials and logarithms · spec MM5.3

Given that $5^x = 2$ and $2^y = 125$, find the value of xy .

Answer: B

$5^{xy} = (5^x)^y = 2^y = 125 = 5^3$, so $xy = 3$.

Question 5 LOG-14 · Exponentials and logarithms · spec MM5.3

The number of integers n satisfying $2^n < 1000 < 2^{n+3}$ is

Answer: A

$2^9 = 512 < 1000 < 1024 = 2^{10}$. Need $n \leq 9$ and $n + 3 \geq 10$, so $n = 7, 8, 9$: three integers.

Question 6 LOG-17 · Exponentials and logarithms · spec MM5.3

The equation $9^x - k \cdot 3^x + 9 = 0$ has exactly one real solution. Find the value of k .

Answer: E

With $y = 3^x > 0$: $y^2 - ky + 9 = 0$. Exactly one solution for x means exactly one positive root: a repeated root ($k^2 = 36$, product $9 > 0$ so the roots have the sign of k): $k = 6$ (repeated root $y = 3$, $x = 1$). $k = -6$ gives $y = -3$, no solutions. (Two distinct positive roots would give two solutions.)

Question 7 LOG-19 · Exponentials and logarithms · spec MM5.3

A quantity halves every 5 hours. Which expression gives the number of hours for it to fall to $\frac{1}{10}$ of its initial value?

Answer: D

$(\frac{1}{2})^{t/5} = \frac{1}{10}$. Taking logs to base 10: $-\frac{t}{5}\log_{10} 2 = -1$, so $t = \frac{5}{\log_{10} 2}$ (about 16.6 hours).

Question 8 X2-06 · Exponentials and logarithms · spec MM5.2

Which of the following is a counterexample to the claim ****for all positive x and y ,**
 $\log_{10}(x + y) \leq \log_{10} x + \log_{10} y$ ******?

Answer: E

$x = y = 1$: $\log 2 > 0 = \log 1 + \log 1$, so the inequality fails. For the others $x + y \leq xy$ and the claim holds (log is increasing).

Question 9 X2-22 · Exponentials and logarithms · spec MM5.3

Consider the statement: ****for all real x , $2^x > x^2$.**** Which of the following is a counterexample?

Answer: B

$2^3 = 8 < 9 = 3^2$, so $x = 3$ breaks the claim. The other options satisfy it: $2 > 1$, $32 > 25$, $1 > 0$, $1024 > 100$.